

Skip List
–Display List
–Program Analysis

Program:

```
// Display the skip list layer by layer
void displayList(struct SkipList *lst) {
    cout << "\n--- Skip List ---" << endl;
    for (int i = 0; i <= lst->level; i++) {
        struct Node *node = lst->header->forward[i];
        cout << "Level " << i << ": ";
        while (node != NULL) {
            cout << node->key << " ";
            node = node->forward[i];
        }
        cout << endl;
    }
    cout << "-----" << endl;
}

case 4:
    displayList(list);
    break;
```

Program Analysis

```
displayList (list);
```

displayList(list: ff08) → passed.

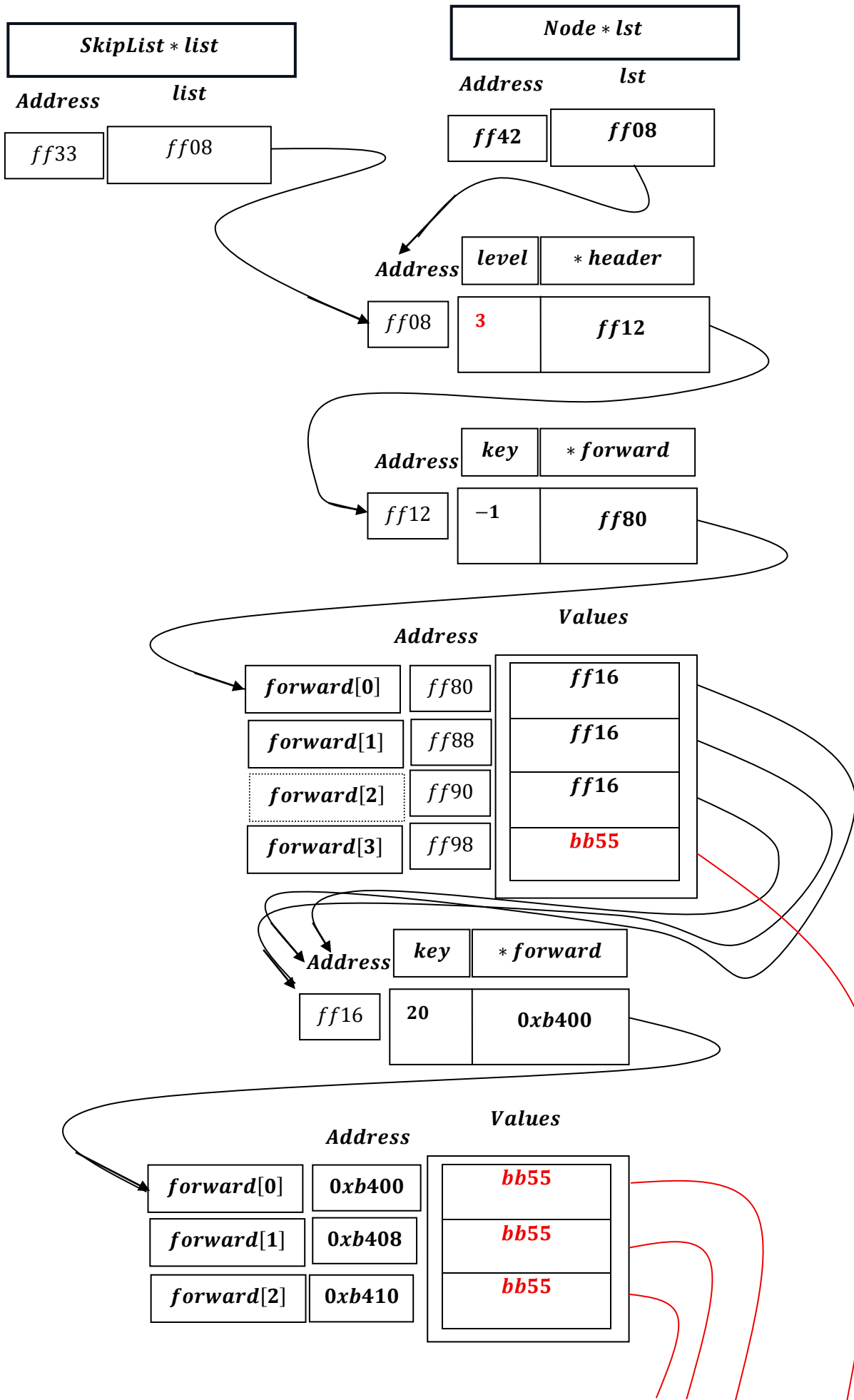
And,

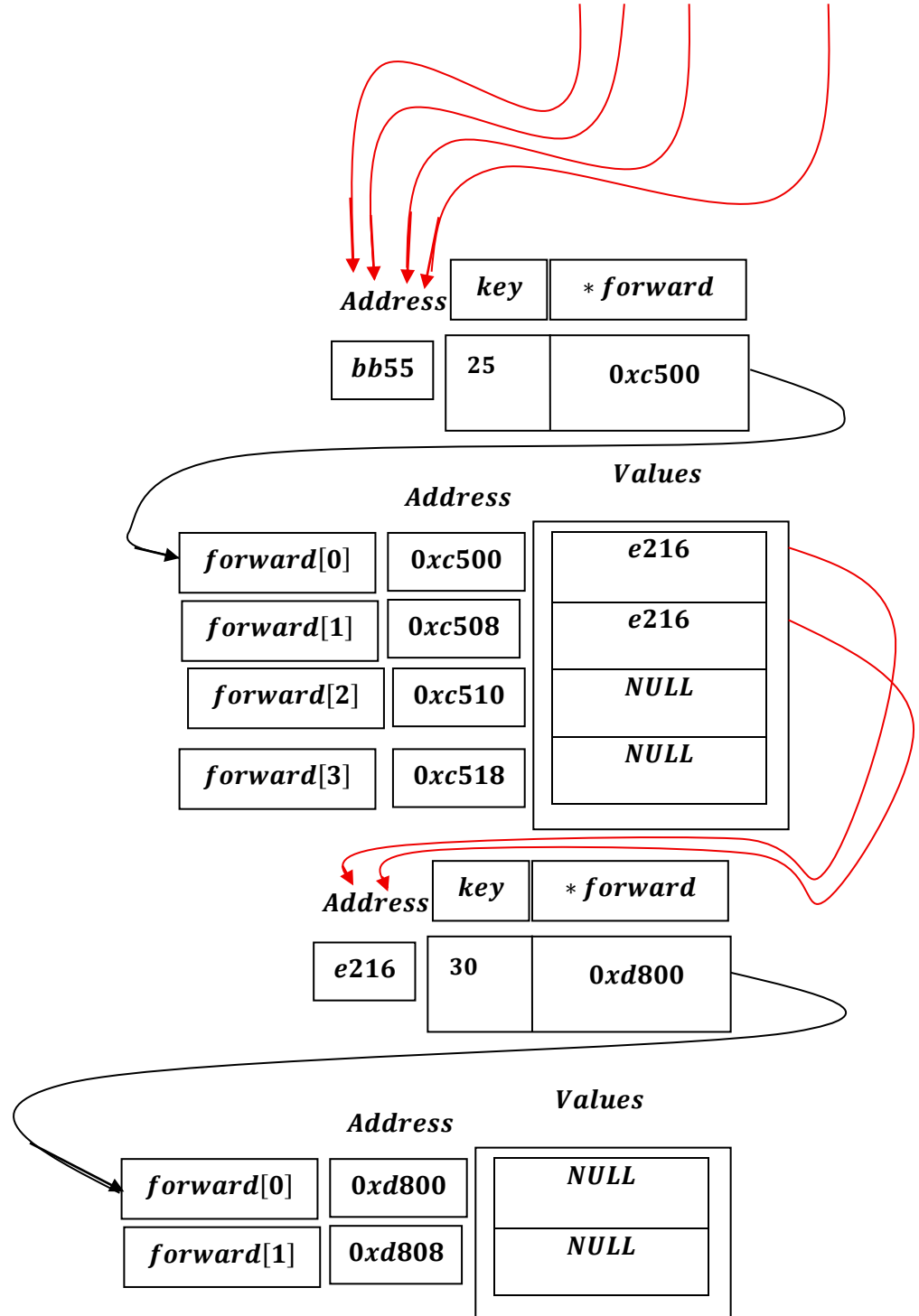
```
void displayList(struct SkipList *lst)
```

displayList(list := lst: ff08) → Assigned

```
cout << "\n--- Skip List ---" << endl;
```

Print: ---Skip List---





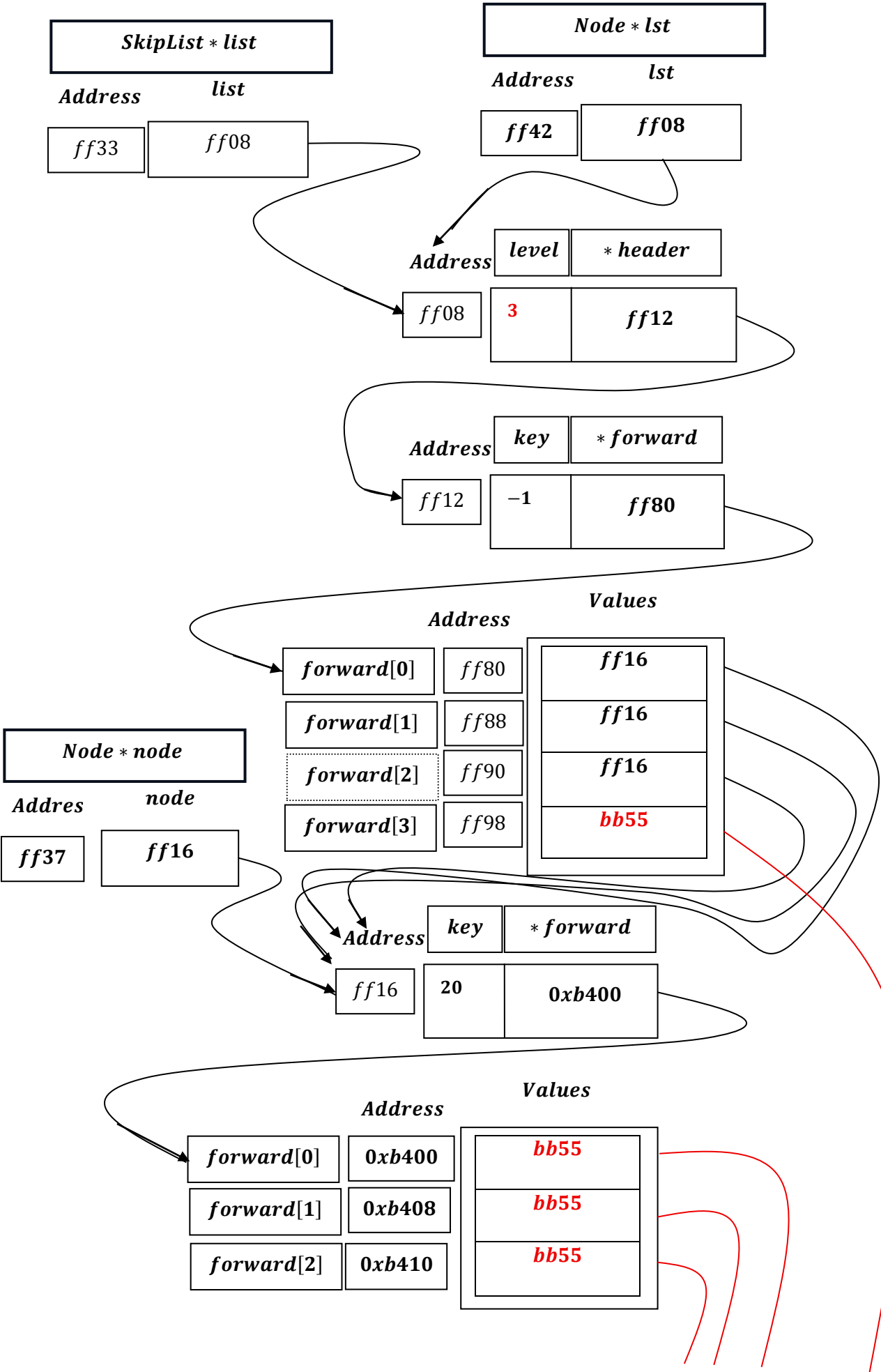
```

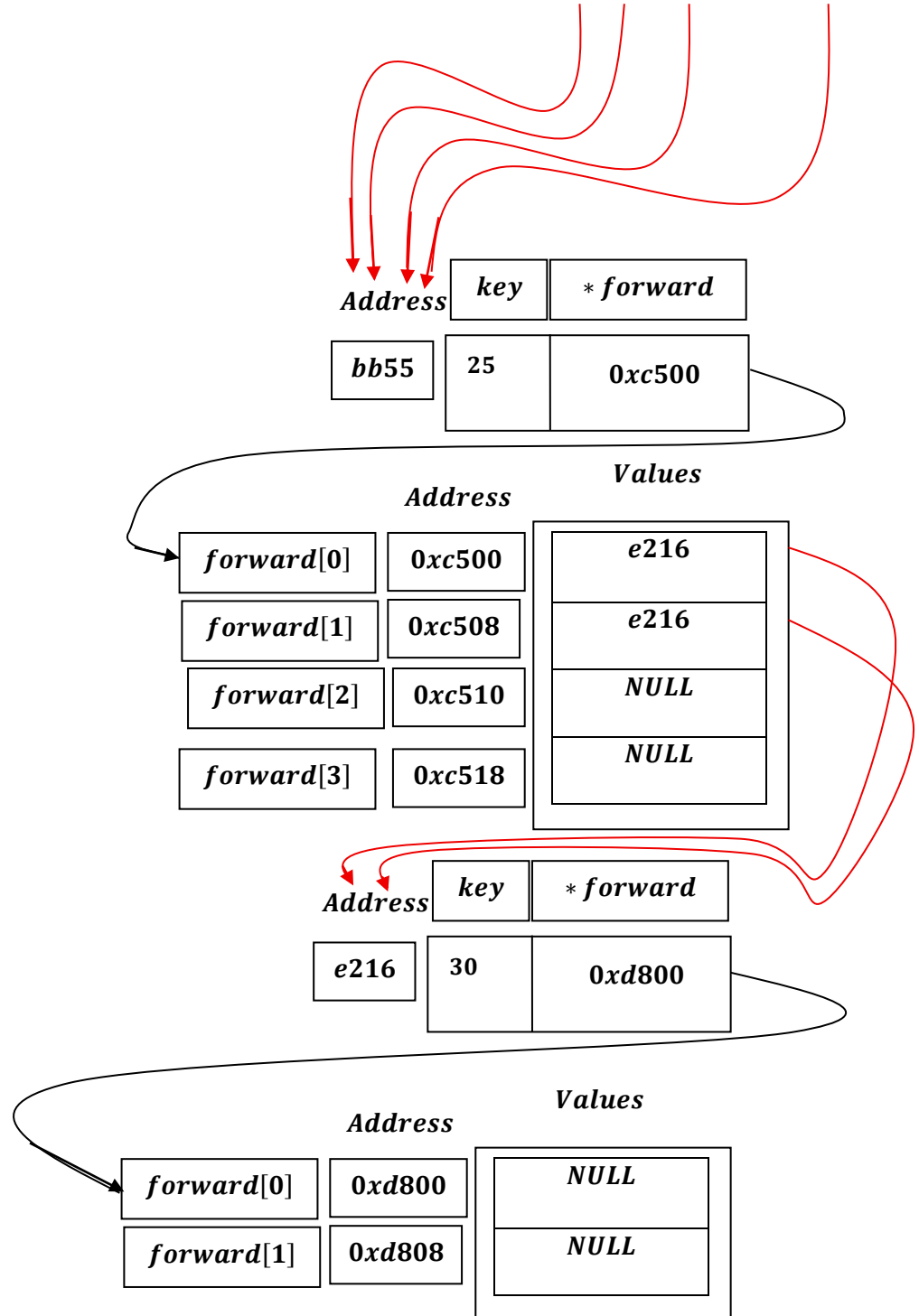
for (int i = 0; i <= lst->level; i++) {
    struct Node *node = lst->header->forward[i];
    cout << "Level " << i << ": ";
    while (node != NULL) {
        cout << node->key << " ";
        node = node->forward[i];
    }
    cout << endl;
}
cout << "-----" << endl;
}

```

for: int i = 0; i ≤ lst → level: 3[True]

*struct Node * node = lst: ff08 → header: ff12 → forward[0]: ff16;*





```
for: int i = 0; i ≤ lst → level: 3[True]
    struct Node * node = lst: ff08 → header: ff12 → forward[0]: ff16;
```

Print : `` Level ``0 ``:

```
while (node != NULL) {
    cout << node->key << " ";
    node = node->forward[i];
}
```

node ≠ *NULL*[*True*]

Print: ``node: ff16 → key" : 20

node = node: ff16 → forward[0]: bb55

node ≠ *NULL*[*True*]

Print: ``node: bb55 → key" : 25

node = node: bb55 → forward[0]: e216

node ≠ *NULL*[*True*]

Print: ``node: e216 → key" : 30

node = node: e216 → forward[0]: NULL

node = *NULL*[*False*]

Final Print:

Level 0 : 20 25 30

```
cout << endl;
```

Next Line of code

$i++ = i + 1 = 0 + 1 = 1$

for: int i = 1; i ≤ lst → level: 3[True]

struct Node * node = lst: ff08 → header: ff12 → forward[1]: ff16;

Print : `` Level ``1 ``:

```
while (node != NULL) {  
    cout << node->key << " ";  
    node = node->forward[i];  
}
```

node ≠ NULL[True]

Print: ``node: ff16 → key" : 20

node = node: ff16 → forward[1]: bb55

node ≠ NULL[True]

Print: ``node: bb55 → key" : 25

node = node: bb55 → forward[1]: e216

node ≠ NULL[True]

Print: ``node: e216 → key" : 30

node = node: e216 → forward[1]: NULL

node = NULL[False]

Final Print:

Level 0 : 20 25 30

Level 1 : 20 25 30

```
cout << endl;
```

Next Line of code

$i++ = i + 1 = 1 + 1 = 2$

for: int i = 2; i ≤ lst → level: 3[True]

struct Node * node = lst: ff08 → header: ff12 → forward[2]: ff16;

Print : `` Level ``2 ``:``

```
while (node != NULL) {  
    cout << node->key << " ";  
    node = node->forward[i];  
}
```

node ≠ NULL[True]

Print: ``node: ff16 → key" : 20

node = node: ff16 → forward[2]: bb55

node ≠ NULL[True]

Print: ``node: bb55 → key" : 25

node = node: bb55 → forward[2]: NULL

node = NULL[False]

Final Print:

Level 0 : 20 25 30

Level 1 : 20 25 30

Level 2 : 20 25

```
cout << endl;
```

Next Line of code

$i++ = i + 1 = 2 + 1 = 3$

for: int i = 3; i ≤ lst → level: 3[True]

struct Node * node = lst: ff08 → header: ff12 → forward[3]: bb55;

Print : `` Level ``3 ``:``

```
while (node != NULL) {  
    cout << node->key << " ";  
    node = node->forward[i];  
}
```

node ≠ NULL[True]

Print: ``node: bb55 → key" : 30

node = node: bb55 → forward[3]: NULL

node = NULL[False]

Final Print:

Level 0 : 20 25 30

Level 1 : 20 25 30

Level 2 : 20 25

Level 3 : 30

```
cout << endl;
```

Next Line of code

i ++ = i + 1 = 3 + 1 = 4

for: int i = 3; i ≤ lst → level: 3 ***False***
for
Loop
Exit

∴ Final Print:

Level 0 : 20 25 30

Level 1 : 20 25 30

Level 2 : 20 25

Level 3 : 30
